

Aerobic Oxidation of Propargylic Alcohols to α,β -Unsaturated Alkynals or Alkynones Catalyzed by $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO and Sodium Chloride in Toluene

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Abstract: A practical aerobic oxidation of propargylic alcohols using $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO and sodium chloride in toluene at room temperature was applied to various type of propargylic alcohols affording α,β -unsaturated alkynals or alkynones in good to excellent yields. This protocol could be applied in academic laboratories as well as in industrial-scale production.

Key words: aerobic oxidation, propargylic alcohols, catalysis, iron nitrate, TEMPO

As important intermediates in organic synthesis, α,β -unsaturated alkynals and alkynones are very useful precursors for the preparation of various aromatic compounds,¹ heterocyclic compounds,² α,β -enones,³ β -keto-1,3-dithianes,⁴ $\alpha,\beta,\gamma,\delta$ -dienones,⁵ chiral propargylic alcohols⁶ as well as other compounds.⁷ Many approaches for the synthesis of α,β -unsaturated alkynals or alkynones such as acylation of alkynyl organometallic reagents⁸ or alkynylboronates,⁹ cross-coupling between terminal alkynes and acyl chlorides,¹⁰ and carbonylative Sonogashira coupling reaction have been reported.¹¹ The most convenient way to prepare such compounds should be the oxidation of its corresponding propargylic alcohols. Various recipes for such purpose using stoichiometric oxidants such as MnO_2 ,¹² PCC/PDC,¹³ Swern oxidation,¹⁴ and hypervalent iodine compounds¹⁵ have so far been reported in literature. However, the reaction would yield almost the same amount of oxidant-derived wastes, thus, causing heavy burden to the environment. Therefore, development of a new catalytic method using a clean oxidant under mild reaction conditions suitable for industrial production is highly desirable.

In the last decade, many aerobic oxidation examples were developed using copper, vanadium, ruthenium, or cobalt complex as catalysts for primary and secondly propargylic alcohols. Ishii and co-workers¹⁶ reported the first example of aerobic oxidation of secondly propargylic alcohol using $\text{Cu}(\text{acac})_2\text{-NHPI}$ (*N*-hydroxyphthalimide) in acetonitrile at 70 °C, but the conversion was low. Uemura¹⁷ and

Hanson¹⁸ reported two different aerobic catalytic oxidation using vanadium complex as catalyst at 80–100 °C for primary and secondly propargylic alcohols; Sain¹⁹ developed a cobalt phthalocyanine complex catalyzed aerobic oxidation in xylene under reflux while Pedro²⁰ reported another cobalt *o*-phenylenebis(*N*-methyloxamidate) complex catalyzed aerobic oxidation in acetonitrile at room temperature for secondly propargylic alcohols independently; and Katsuki and co-workers²¹ reported a kinetic resolution of *sec*-alcohols including *sec*-propargylic alcohol using a Ru complex as catalyst through catalytic aerobic oxidation. However, heating or precious metal catalyst was required in these protocols.

Recently, our group²² has reported a general and practical approach to aerobic oxidation of alcohols bearing a wide range of substrates including propargylic alcohols at room temperature using $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO and sodium chloride as catalyst in 1,2-dichloroethane, yielding aldehydes or ketones in good to excellent yields. However, such a chlorinated solvent is disfavored for large scale industrial application. In this paper, we wish to report an industrially practical protocol for such a purpose by using toluene as the solvent.

The solvent effect of the aerobic oxidation reaction at room temperature was first examined in order to select an industrial-friendly solvent with 1-phenyl-3-(trimethylsilyl)prop-2-yn-1-ol (**1a**) as the substrate. The results of the oxidation of this substrate in the presence of 5 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 3 mol% TEMPO, and 5 mol% NaCl under the atmospheric pressure of oxygen in different solvents are summarized in Table 1. We were delighted to find that when using toluene, *p*-xylene, acetonitrile, and 1,2-dichloroethane (DCE) as the solvent, the substrate could be converted into its corresponding ynone **2a** within two hours in almost quantitative yield (Table 1, entries 1, 2, 4, and 5). As a cheap solvent, toluene is widely used for industrial production.²³ Therefore, 5 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 3 mol% TEMPO, and 5 mol% NaCl in toluene were chosen as the standard reaction conditions.

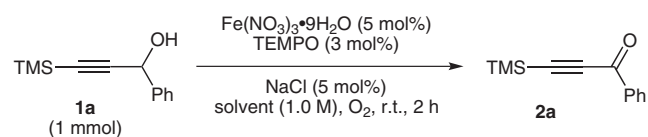
The scope of substrates for the aerobic oxidation of propargylic alcohols under the optimized conditions was examined. First, different substituents in the aromatic ring

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Table 1 Room-Temperature $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ -Catalyzed Aerobic Oxidation of Propargylic Alcohol **1a** to Ketone **2a** in Different Solvents

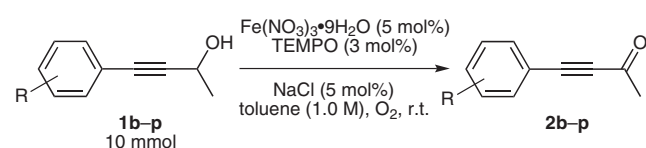
Entry	Solvent	Yield of 2a (%) ^a
1	toluene	99
2	<i>p</i> -xylene	99
3	acetone	11(84) ^b
4	MeCN	99
5	DCE	98

^a Yield determined by ¹H NMR analysis.^b The yield was 11% and the recovery of alcohol was 84%.

of 4-aryl-3-propargylic alcohols were examined (Table 2). A wide range of functional groups such as F, Cl, Br, I, CO₂Et, CN, NO₂, NHTs, alkyl, MeO, etc. were smoothly tolerated. When R = *m*-F, *o*-F, *p*-Cl, *p*-CO₂Et, and *p*-NO₂, the substrates were efficiently oxidized to give the corresponding ketones in high yields with 5 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 3 mol% TEMPO, and 5 mol% NaCl (Table 2, entries 5, 6, 7, 12, and 15); when R = H, *p*-F, *p*-Br, *p*-I, *p*-CN, *p*-NHTs, the substrates were oxidized to give the corresponding ketones in full conversion with 5 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 5 mol% TEMPO, and 10 mol% NaCl (Table 2, entries 2, 4, 9, 11, 14, and 17). Compared to the electron-withdrawing group, when an electron-donating group was present in the aromatic ring, the reaction was obviously slower. When R = *o*-MeO, *m*-MeO, and *p*-MeO, the catalyst loading should be increased to 10 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 10 mol% TEMPO, and 10 mol% NaCl for full conversion of the substrate (Table 2, entries 20, 22, and 23).

The optimized conditions were next applied to 1-substituted or unsubstituted undec-2-yn-1-ols (Table 3). When R = aryl, the reaction could reach completion in 7.1 hours (Table 3, entries 1–5). A slight difference in reaction time was noted when changing the position of the substituent in the aryl ring (Table 3, entries 3–5). When R = H, the reaction could reach completion in five hours (Table 3, entry 7); however, when R = alkyl, the reaction time increased to 10.3 hours (Table 3, entry 6). In all the cases, the yields were excellent.

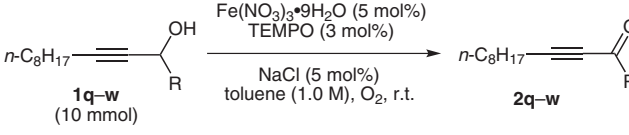
Further study indicates that the scope is very broad. When R¹ = Ph and R² = Ph or H, the reaction could reach completion within two hours (Table 4, entries 1 and 2); with R¹ = aryl and R² = alkyl, the reaction was much slower, although the yield was still excellent (Table 4, entry 3). The optimized reaction conditions were also applied to terminal propargylic alcohols (Table 4, entries 4 and 5). The TMS-substituted propargylic alcohols also worked smoothly (Table 4, entries 7 and 8).

Table 2 Room-Temperature Aerobic Oxidation of 4-Aryl-3-propargylic Alcohols to the Corresponding Ketones^a

Entry	Substrate	R	Method	Time (h)	Isolated yield of 2 (%)
1	1b	H	A	24	77 (2b) ^b
2	1b	H	B	2	91 (2b)
3	1c	<i>p</i> -F	A	24	58 (2c) ^b
4	1c	<i>p</i> -F	B	2	90 (2c)
5	1d	<i>m</i> -F	A	12	83 (2d)
6	1e	<i>o</i> -F	A	4.5	94 (2e)
7	1f	<i>p</i> -Cl	A	9.5	86 (2f)
8	1g	<i>p</i> -Br	A	24	75 (2g) ^b
9	1g	<i>p</i> -Br	B	3.3	93 (2g)
10	1h	<i>p</i> -I	A	24	59 (2h) ^b
11	1h	<i>p</i> -I	B	2	82 (2h)
12	1i	<i>p</i> -CO ₂ Et	A	24	79 (2i)
13	1j	<i>p</i> -CN	A	24	65 (11) (2j) ^c
14	1j	<i>p</i> -CN	B	2.7	87 (2j)
15	1k	<i>p</i> -NO ₂	A	12.4	92 (2k)
16	1l	<i>p</i> -NHTs	A	46	61 (2l) ^d
17	1l	<i>p</i> -NHTs	B	2	84 (2l)
18	1m	<i>p</i> -Me	A	24	36 (41) (2m) ^c
19	1m	<i>p</i> -Me	B	2.1	91 (2m)
20	1n	<i>p</i> -MeO	C	4.2	70 (2n)
21	1o	<i>m</i> -MeO	A	24	55 (27) (2o) ^c
22	1o	<i>m</i> -MeO	C	1.8	91 (2o)
23	1p	<i>o</i> -MeO	C	1.8	87 (2p)

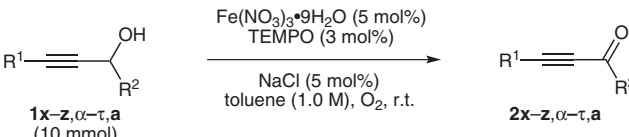
^a Method A: substrate (10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (5 mol%), TEMPO (3 mol%), and NaCl (5 mol%) in toluene (10 mL); Method B: substrate (10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (5 mol%), TEMPO (5 mol%), and NaCl (10 mol%) in toluene (10 mL); Method C: substrate (10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (10 mol%), TEMPO (10 mol%), and NaCl (10 mol%) in toluene (10 mL).

^b Incomplete conversion as monitored by TLC after 24 h.^c Isolated yield with the recovery of starting alcohol is shown in the parentheses.^d The yield was determined by ¹H NMR analysis and 24% of starting material (propargylic alcohol) was recovered.

Table 3 Room-Temperature Aerobic Oxidation of 1-Substituted/Unsubstituted Undec-2-yn-1-ols to the Corresponding Aldehydes or Ketones


Entry	Substrate	R	Time (h)	Isolated yield of 2 (%)
1	1q	<i>p</i> -ClC ₆ H ₄	2.5	98 (2q)
2	1r	<i>p</i> -CF ₃ C ₆ H ₄	2.5	98 (2r)
3	1s	<i>p</i> -MeOC ₆ H ₄	2.5	92 (2s)
4	1t	<i>m</i> -MeOC ₆ H ₄	3.3	99 (2t)
5	1u	<i>o</i> -MeOC ₆ H ₄	7.1	96 (2u)
6	1v	<i>n</i> -C ₅ H ₁₁	10.3	98 (2v)
7	1w	H	5	97 (2w)

^a Standard reaction conditions: substrate (10 mmol), Fe(NO₃)₃·9H₂O (5 mol%), TEMPO (3 mol%), and NaCl (5 mol%) in toluene (10 mL).

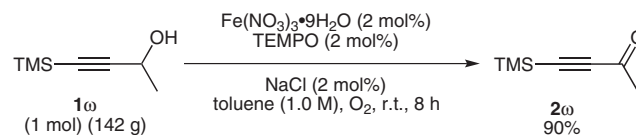
Table 4 Room-Temperature Aerobic Oxidation of Differently Substituted Propargylic Alcohols to the Corresponding Aldehydes or Ketones


Entry	Substrate	R ¹	R ²	Time (h)	Isolated yield of 2 (%)
1	1x	Ph	H	2	94 (2x)
2	1y	Ph	Ph	1.8	99 (2y)
3	1z	Ph	<i>n</i> -C ₄ H ₉	16.5	93 (2z)
4	1α	H	Ph	6.8	92 (2α)
5	1β	H	<i>n</i> -C ₉ H ₁₉	13.2	95 (2β)
6	1γ	<i>n</i> -C ₄ H ₉	<i>p</i> -CF ₃ C ₆ H ₄	4	87 (2γ)
7	1τ	TMS	<i>n</i> -C ₅ H ₁₁	11.5	97 (2τ)
8	1a	TMS	Ph	3.5	98 (2a)

^a Standard reaction conditions: of substrate (10 mmol), Fe(NO₃)₃·9H₂O (5 mol%), TEMPO (3 mol%), and NaCl (5 mol%) in toluene (10 mL).

A Large-Scale Reaction for Practical Use

To further show the practicality and efficiency of this catalytic system, a 1.0 mol reaction of 4-(trimethylsilyl)but-3-yn-2-ol was conducted by using 2 mol% each of Fe(NO₃)₃·9H₂O, TEMPO, and NaCl to give 4-(trimethylsilyl)but-3-yn-2-one (**2ω**) in 90% isolated yield in eight hours by simple distillation (Scheme 1).

**Scheme 1**

In summary, we have developed a practical aerobic oxidation protocol for various types of propargylic alcohols using Fe(NO₃)₃·9H₂O/TEMPO/NaCl in toluene at room temperature affording the α,β-unsaturated alkynal or alkynones in good to excellent yields. This eco-friendly oxidation system may be applied in academic laboratories as well as industrial scale production.

¹H and ¹³C NMR spectra were recorded with an instrument operating at 300 MHz for ¹H NMR and 75 MHz for ¹³C NMR in CDCl₃. Chemical shifts (δ) are given in parts per million (ppm). Infrared spectra were recorded on an FTIR spectrometer. MS and HRMS data were collected in EI mode. Compounds **1b–p** and **1x** were synthesized by Sonogashira coupling²⁴ and all the other starting materials were synthesized according to the literature.²⁵ Other reagents were commercially available. Petroleum ether (PE) used refers to the fraction boiling in the range 60–90 °C.

Oxidation of Propargylic Alcohols; General Procedure

To a 50 mL three-necked flask were added Fe(NO₃)₃·9H₂O (0.5 mmol), TEMPO (0.30 mmol), NaCl (0.50 mmol), and toluene (10 mL). The propargylic alcohol (10 mmol) was then added to the suspension and the resulting mixture was placed under the atmosphere of O₂ from a balloon and stirred at r.t. until the reaction was complete as monitored by TLC (eluent: PE–EtOAc, 10:1). The resulting mixture was purified by column chromatography on silica gel [PE: 200 mL, PE–Et₂O (10:1): 300–400 mL] to afford the corresponding aldehyde or ketone. The reagent amounts used for methods A–C are given below (cf. Table 2).

Method A: Alcohol (10 mmol), Fe(NO₃)₃·9H₂O (5 mol%), TEMPO (3 mol%), and NaCl (5 mol%) in toluene (10 mL).

Method B: Alcohol (10 mmol), Fe(NO₃)₃·9H₂O (5 mol%), TEMPO (5 mol%), and NaCl (10 mol%) in toluene (10 mL).

Method C: Alcohol (10 mmol), Fe(NO₃)₃·9H₂O (10 mol%), TEMPO (10 mol%), and NaCl (10 mol%) in toluene (10 mL).

4-Phenylbut-3-yn-2-one (**2b**)

Method A: The reaction of 4-phenylbut-3-yn-2-ol (1.4420 g, 10 mmol), Fe(NO₃)₃·9H₂O (205.0 mg, 0.5 mmol), TEMPO (47.4 mg, 0.3 mmol), and NaCl (29.0 mg, 0.5 mmol) in toluene (10 mL) afforded **2b** (1.0926 g, 77%) as an oil.

Method B: The reaction of 4-phenylbut-3-yn-2-ol (1.4620 g, 10 mmol), Fe(NO₃)₃·9H₂O (202.3 mg, 0.5 mmol), TEMPO (78.4 mg, 0.5 mmol), and NaCl (58.5 mg, 1.0 mmol) in toluene (10 mL) afforded **2b**²⁶ (1.3131 g, 91%) as an oil.

IR (neat): 2202, 1672, 1490, 1444, 1415, 1359, 1325, 1280, 1157, 1067, 1025, 977 cm^{−1}.

^1H NMR (300 MHz, CDCl_3): δ = 7.61–7.54 (m, 2 H, ArH), 7.50–7.34 (m, 3 H, ArH), 2.45 (s, 3 H, COCH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.59, 133.04, 130.72, 128.62, 119.89, 90.18, 88.25, 32.72.

MS (EI): m/z (%) = 144 (M^+ , 20.08), 129 (100).

4-(4-Fluorophenyl)but-3-yn-2-one (2c)

Method A: The reaction of 4-(4-fluorophenyl)but-3-yn-2-ol (1.6410 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (200.6 mg, 0.5 mmol), TEMPO (47.3 mg, 0.3 mmol), and NaCl (28.2 mg, 0.5 mmol) in toluene (10 mL) afforded **2c** (0.9348 g, 58%) as an oil.

Method B: The reaction of 4-(4-fluorophenyl)but-3-yn-2-ol (1.6419 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.2 mg, 0.5 mmol), TEMPO (78.0 mg, 0.5 mmol), and NaCl (58.4 mg, 1.0 mmol) in toluene (10 mL) afforded **2c**²⁷ (1.4523 g, 90%) as an oil.

IR (neat): 2204, 1673, 1600, 1505, 1416, 1358, 1283, 1231, 1152, 1095, 1016, 977 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.61–7.53 (m, 2 H, ArH), 7.12–7.03 (m, 2 H, ArH), 2.44 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.29, 163.80 (d, J = 254.0 Hz), 135.17 (d, J = 8.7 Hz), 116.01 (d, J = 22.5 Hz), 115.81, 89.024, 87.98 (d, J = 1.1 Hz), 32.49.

^{19}F NMR (282 MHz, CDCl_3): δ = –106.24.

MS (EI): m/z (%) = 162 (M^+ , 17.88), 147 (100).

4-(3-Fluorophenyl)but-3-yn-2-one (2d)

Method A: The reaction of 4-(3-fluorophenyl)but-3-yn-2-ol (1.6297 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (205.1 mg, 0.5 mmol), TEMPO (46.8 mg, 0.3 mmol), and NaCl (29.4 mg, 0.5 mmol) in toluene (10 mL) afforded **2d** (1.3378 g, 83%) as an oil.

IR (neat): 2201, 1671, 1608, 1580, 1486, 1431, 1359, 1294, 1266, 1200, 1160, 1132, 1078, 1020, 982 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.41–7.32 (m, 2 H, ArH), 7.30–7.23 (m, 1 H, ArH), 7.23–7.12 (m, 1 H, ArH), 2.46 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.30, 162.19 (d, J = 248.0 Hz), 130.33 (d, J = 8.3 Hz), 128.85 (d, J = 3.2 Hz), 121.66 (d, J = 9.3 Hz), 119.59 (d, J = 23.1 Hz), 118.13 (d, J = 21.5 Hz), 88.38, 88.24 (d, J = 3.2 Hz), 32.67.

^{19}F NMR (282 MHz, CDCl_3): δ = –111.73.

MS (EI): m/z (%) = 162 (M^+ , 20.82), 147 (100).

HRMS: m/z calcd for $\text{C}_{10}\text{H}_7\text{FO}$ (M^+): 162.0481; found: 162.0482.

4-(2-Fluorophenyl)but-3-yn-2-one (2e)

Method A: The reaction of 4-(2-fluorophenyl)but-3-yn-2-ol (1.6417 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.3 mg, 0.5 mmol), TEMPO (46.9 mg, 0.3 mmol), and NaCl (29.0 mg, 0.5 mmol) in toluene (10 mL) afforded **2e** (1.5243 g, 94%) as an oil.

IR (neat): 2209, 1676, 1610, 1578, 1491, 1450, 1418, 1359, 1297, 1263, 1221, 1200, 1166, 1151, 1103, 1020, 978 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.57–7.39 (m, 2 H, ArH), 7.20–7.07 (m, 2 H, ArH), 2.46 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.18, 163.41 (d, J = 255.1 Hz), 134.53, 132.69 (d, J = 8.2 Hz), 124.23 (d, J = 3.8 Hz), 115.77 (d, J = 20.9 Hz), 108.65 (d, J = 15.4 Hz), 92.41 (d, J = 3.3 Hz), 83.39, 32.60.

^{19}F NMR (282 MHz, CDCl_3): δ = –107.26.

MS (EI): m/z (%) = 162 (M^+ , 21.31), 147 (100).

HRMS: m/z calcd for $\text{C}_{10}\text{H}_7\text{FO}$ (M^+): 162.0481; found: 162.0480.

4-(4-Chlorophenyl)but-3-yn-2-one (2f)

Method A: The reaction of 4-(4-chlorophenyl)but-3-yn-2-ol (1.8164 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (201.8 mg, 0.5 mmol), TEMPO (47.3

mg, 0.3 mmol), and NaCl (28.7 mg, 0.5 mmol) in toluene (10 mL) afforded **2f**²⁸ (1.5386 g, 86%) as a solid; mp 54–55 °C (*c*-hexane).

IR (neat): 2203, 1671, 1582, 1486, 1417, 1401, 1359, 1278, 1203, 1156, 1132, 1088, 1014, 983 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.50 (dt, J = 8.7, 2.1 Hz, 2 H, ArH), 7.36 (dt, J = 8.4, 2.1 Hz, 2 H, ArH), 2.45 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.36, 137.13, 134.20, 129.08, 118.33, 88.87, 88.78, 32.67.

MS (EI): m/z (%) 180 [M^+ (^{37}Cl), 9.05], 178 [M^+ (^{35}Cl), 27.38], 163 (100).

4-(4-Bromophenyl)but-3-yn-2-one (2g)

Method A: The reaction of 4-(4-bromophenyl)but-3-yn-2-ol (2.2507 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.0 mg, 0.5 mmol), TEMPO (47.2 mg, 0.3 mmol), and NaCl (29.5 mg, 0.5 mmol) in toluene (10 mL) afforded **2g** (1.6671 g, 75%) as a solid.

Method B: The reaction of 4-(4-bromophenyl)but-3-yn-2-ol (2.2511 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.3 mg, 0.5 mmol), TEMPO (78.4 mg, 0.5 mmol), and NaCl (58.7 mg, 1.0 mmol) in toluene (10 mL) afforded **2g**²⁸ (2.0676 g, 93%) as a solid; mp 64–65 °C (*c*-hexane).

IR (neat): 2201, 1668, 1597, 1585, 1506, 1490, 1465, 1435, 1419, 1394, 1358, 1277, 1246, 1233, 1154, 1113, 1096, 1069, 1046, 1011, 977 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.52 (dt, J = 9.0, 2.0 Hz, 2 H, ArH), 7.41 (dt, J = 8.7, 2.0 Hz, 2 H, ArH), 2.44 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.35, 134.27, 131.99, 125.50, 118.76, 88.94, 88.80, 32.65.

MS (EI): m/z (%) = 224 [M^+ (^{81}Br), 22.67], 222 [M^+ (^{79}Br), 23.47], 207 (100).

4-(4-Iodophenyl)but-3-yn-2-one (2h)

Method A: The reaction of 4-(4-iodophenyl)but-3-yn-2-ol (2.7168 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (201.2 mg, 0.5 mmol), TEMPO (47.1 mg, 0.3 mmol), and NaCl (28.9 mg, 0.5 mmol) in toluene (10 mL) afforded **2h** (1.5913 g, 59%) as a solid; mp 83–84 °C (*c*-hexane).

Method B: The reaction of 4-(4-iodophenyl)but-3-yn-2-ol (2.7266 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (206.6 mg, 0.5 mmol), TEMPO (78.5 mg, 0.5 mmol), and NaCl (57.0 mg, 1.0 mmol) in toluene (10 mL) afforded **2h** (2.2059 g, 82%) as a solid.

IR (neat): 3000, 2945, 2889, 2834, 2201, 1709, 1680, 1654, 1579, 1482, 1420, 1391, 1355, 1279, 1154, 1054, 1006, 977 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.74 (dt, J = 8.7, 2.0 Hz, 2 H, ArH), 7.28 (dt, J = 8.7, 1.8 Hz, 2 H, ArH), 2.45 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.29, 137.87, 134.15, 119.27, 97.60, 89.13, 88.91, 32.65.

MS (EI): m/z (%) 270 (M^+ , 49.02), 255 (100).

Anal. Calcd for $\text{C}_{10}\text{H}_7\text{IO}$: C, 44.47; H, 2.61. Found: C, 44.47; H, 2.61.

Ethyl 4-(3-Oxobut-1-ynyl)benzoate (2i)

Method A: The reaction of ethyl 4-(3-hydroxybut-1-ynyl)benzoate (2.1693 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.6 mg, 0.5 mmol), TEMPO (47.5 mg, 0.3 mmol), and NaCl (29.2 mg, 0.5 mmol) in toluene (10 mL) afforded **2i** (1.6899 g, 79%) as a solid; mp 52–53 °C (*c*-hexane).

IR (neat): 2205, 1713, 1669, 1604, 1562, 1507, 1470, 1447, 1404, 1360, 1307, 1272, 1175, 1157, 1105, 1017, 978 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 8.05 (d, J = 8.4 Hz, 2 H, ArH), 7.62 (d, J = 8.4 Hz, 2 H, ArH), 4.39 (q, J = 7.1 Hz, 2 H, OCH_2), 2.47 (s, 3 H, COCH_3), 1.40 (t, J = 7.2 Hz, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 184.29, 165.56, 132.74, 132.05, 129.58, 124.21, 89.73, 88.51, 61.40, 32.71, 14.23.

MS (EI): m/z = 216 (M^+ , 29.16), 201 (100).

Anal. Calcd for $C_{13}H_{12}O_3$: C, 72.21; H, 5.59. Found: C, 72.26; H, 5.53.

4-(3-Oxobut-1-ynyl)benzonitrile (**2j**)

Method A: The reaction of 4-(3-hydroxybut-1-ynyl)benzonitrile (1.7122 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (202.3 mg, 0.5 mmol), TEMPO (46.7 mg, 0.3 mmol), and NaCl (29.4 mg, 0.5 mmol) in toluene (10 mL) afforded **2j** (1.1018 g, 65%) as a solid.

Method B: The reaction of 4-(3-hydroxybut-1-ynyl)benzonitrile (1.7117 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (202.3 mg, 0.5 mmol), TEMPO (78.1 mg, 0.5 mmol), and NaCl (58.7 mg, 1.0 mmol) in toluene (10 mL) afforded **2j** (1.4745 g, 87%) as a solid; mp 106–107 °C (*c*-hexane).

IR (neat): 2227, 2206, 1671, 1499, 1408, 1359, 1277, 1186, 1152, 1018, 981 cm^{-1} .

1H NMR (300 MHz, $CDCl_3$): δ = 7.71–7.62 (m, 4 H, ArH), 2.47 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 183.91, 133.17, 132.19, 124.65, 117.79, 113.92, 90.50, 86.78, 32.65.

MS (EI): m/z (%) = 169 (M^+ , 12.91), 154 (100).

Anal. Calcd for $C_{11}H_7NO$: C, 78.09; H, 4.17; N, 8.28. Found: C, 77.89; H, 4.24; N, 8.11.

4-(4-Nitrophenyl)but-3-yn-2-one (**2k**)

Method A: The reaction of 4-(4-nitrophenyl)but-3-yn-2-ol (1.8921 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (201.9 mg, 0.5 mmol), TEMPO (46.6 mg, 0.3 mmol), and NaCl (29.3 mg, 0.5 mmol) in toluene (10 mL) afforded **2k** (1.7159 g, 92%) as a solid (eluent: PE–EtOAc, 10:1); mp 113–114 °C (*c*-hexane).

IR (neat): 2208, 1672, 1597, 1518, 1403, 1340, 1309, 1277, 1213, 1179, 1156, 1109, 1013, 977 cm^{-1} .

1H NMR (300 MHz, $CDCl_3$): δ = 8.24 (d, J = 8.7 Hz, 2 H, ArH), 7.72 (d, J = 8.7 Hz, 2 H, ArH), 2.48 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 183.87, 148.60, 133.59, 126.54, 123.73, 90.96, 86.33, 32.68.

MS (EI) m/z (%) = 189 (M^+ , 18.15), 174 (100).

Anal. Calcd for $C_{10}H_7NO_3$: C, 63.49; H, 3.73; N, 7.40. Found: C, 63.46; H, 3.75; N, 7.28.

4-Methyl-N-[4-(3-oxobut-1-ynyl)phenyl]benzenesulfonamide (**2l**)

Method A: To a 50 mL three-necked flask were added $Fe(NO_3)_3 \cdot 9H_2O$ (203.7 mg, 0.5 mmol), TEMPO (47.5 mg, 0.3 mmol), NaCl (29.5 mg, 0.5 mmol), and toluene (10 mL). *N*-[4-(3-Hydroxybut-1-ynyl)phenyl]-4-methylbenzenesulfonamide (3.1596 g, 10 mmol) was then added to the suspension and the resulting mixture was placed under the atmosphere of O_2 from a balloon and stirred at r.t. for 46 h. The resulting mixture was filtered and concentrated in vacuo to give the crude product. 1H NMR analysis indicated that the yield of **2l** was 61% and 24% of the alcohol was recovered.

Method B: The reaction of *N*-[4-(3-hydroxybut-1-ynyl)phenyl]-4-methylbenzenesulfonamide (3.1484 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (201.4 mg, 0.5 mmol), TEMPO (78.8 mg, 0.5 mmol), and NaCl (58.8 mg, 1.0 mmol) in toluene (10 mL) afforded **2l** (2.6167 g, 84%) as a solid; mp 124–125 °C (*c*-hexane– Et_2O).

IR (neat): 3313, 2988, 2901, 2838, 2203, 1694, 1664, 1604, 1506, 1453, 1401, 1362, 1331, 1282, 1222, 1157, 1087, 1017, 978, 921 cm^{-1} .

1H NMR (300 MHz, $CDCl_3$): δ = 7.74 (d, J = 8.4 Hz, 2 H, ArH), 7.67 (br d, 1 H, NH), 7.43 (d, J = 8.7 Hz, 2 H, ArH), 7.26 (d, J = 7.8 Hz, 2 H, ArH), 7.12 (d, J = 8.7 Hz, 2 H, ArH), 2.43 (s, 3 H, $ArCH_3$), 2.38 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 184.91, 144.48, 139.19, 135.64, 134.45, 129.88, 127.21, 119.56, 115.62, 90.24, 88.56, 32.64, 21.53.

MS (EI): m/z (%) = 313 (M^+ , 100.00).

Anal. Calcd for $C_{17}H_{15}NO_3S$: C, 65.16; H, 4.82; N, 4.47. Found: C, 64.98; H, 4.79; N, 4.20.

4-(*p*-Tolyl)but-3-yn-2-one (**2m**)

Method A: The reaction of 4-(*p*-tolyl)but-3-yn-2-ol (1.6021 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (202.3 mg, 0.5 mmol), TEMPO (46.7 mg, 0.3 mmol), and NaCl (29.5 mg, 0.5 mmol) in toluene (10 mL) afforded **2m** (0.5663 g, 36%) as an oil.

Method B: The reaction of 4-(*p*-tolyl)but-3-yn-2-ol (1.6022 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (202.2 mg, 0.5 mmol), TEMPO (78.3 mg, 0.5 mmol), and NaCl (58.6 mg, 1.0 mmol) in toluene (10 mL) afforded **2m**²⁹ (1.4464 g, 91%) as an oil.

IR (neat): 2224, 1668, 1606, 1509, 1454, 1414, 1357, 1281, 1152, 1063, 1019, 975 cm^{-1} .

1H NMR (300 MHz, $CDCl_3$): δ = 7.47 (d, J = 8.1 Hz, 2 H, ArH), 7.19 (d, J = 7.8 Hz, 2 H, ArH), 2.44 (s, 3 H, $ArCH_3$), 2.38 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 184.51, 141.27, 132.90, 129.24, 116.55, 90.81, 87.95, 32.51, 21.52.

MS (EI): m/z (%) = 158 (M^+ , 22.30), 143 (100).

4-(4-Methoxyphenyl)but-3-yn-2-one (**2n**)

Method C: The reaction of 4-(4-methoxyphenyl)but-3-yn-2-ol (1.7621 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (404.1 mg, 1.0 mmol), TEMPO (156.0 mg, 1.0 mmol), and NaCl (58.4 mg, 1.0 mmol) in toluene (10 mL) afforded **2n**³⁰ (1.2180 g, 70%) as an oil.

IR (neat): 2196, 1665, 1602, 1568, 1509, 1461, 1442, 1419, 1357, 1280, 1252, 1150, 1109, 1026, 976 cm^{-1} .

1H NMR (300 MHz, $CDCl_3$): δ = 7.51 (dt, J = 9.0, 2.4 Hz, 2 H, ArH), 6.88 (dt, J = 8.7, 2.4 Hz, 2 H, ArH), 3.82 (s, 3 H, OCH_3), 2.41 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 184.53, 161.52, 134.97, 114.22, 111.45, 91.38, 88.08, 55.26, 32.47.

MS (EI): m/z (%) = 174 (M^+ , 35.40), 159 (100).

4-(3-Methoxyphenyl)but-3-yn-2-one (**2o**)

Method A: The reaction of 4-(3-methoxyphenyl)but-3-yn-2-ol (1.7618 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (202.3 mg, 0.5 mmol), TEMPO (46.9 mg, 0.3 mmol), and NaCl (29.2 mg, 0.5 mmol) in toluene (10 mL) afforded **2o** (0.9523 g, 55%) as a solid.

Method C: The reaction of 4-(3-methoxyphenyl)but-3-yn-2-ol (1.7618 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (404.3 mg, 1.0 mmol), TEMPO (156.5 mg, 1.0 mmol), and NaCl (58.4 mg, 1.0 mmol) in toluene (10 mL) afforded **2o** (1.5865 g, 91%) as a solid; mp 54–55 °C (*c*-hexane).

IR (neat): 2195, 1668, 1602, 1573, 1510, 1483, 1465, 1425, 1356, 1321, 1284, 1254, 1221, 1185, 1153, 1093, 1041, 980 cm^{-1} .

1H NMR (300 MHz, $CDCl_3$): δ = 7.31 (t, J = 8.1 Hz, 1 H, ArH), 7.17 (dt, J = 7.5, 1.2 Hz, 1 H, ArH), 7.10–7.06 (m, 1 H, ArH), 7.15 (ddd, J = 8.4, 2.7, 0.9 Hz, 1 H, ArH), 3.82 (s, 3 H, OCH_3), 2.46 (s, 3 H, CH_3).

^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 184.56, 159.33, 129.67, 125.48, 120.73, 117.47, 117.45, 90.15, 87.86, 55.29, 32.65.

MS (EI): m/z (%) = 174 (M^+ , 34.12), 159 (100).

Anal. Calcd for $C_{11}H_{10}O_2$: C, 75.84; H, 5.79. Found: C, 75.94; H, 5.78.

4-(2-Methoxyphenyl)but-3-yn-2-one (**2p**)

Method C: The reaction of 4-(2-methoxyphenyl)but-3-yn-2-ol (1.7624 g, 10 mmol), $Fe(NO_3)_3 \cdot 9H_2O$ (404.2 mg, 1.0 mmol),

TEMPO (156.5 mg, 1.0 mmol), and NaCl (58.3 mg, 1.0 mmol) in toluene (10 mL) afforded **2p** (1.5166 g, 87%) as a solid; mp 52–53 °C (*c*-hexane).

IR (neat): 2199, 1668, 1595, 1574, 1538, 1491, 1462, 1434, 1362, 1296, 1278, 1246, 1184, 1159, 1110, 1041, 1018, 979 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.49 (dd, *J* = 7.7, 1.7 Hz, 1 H, ArH), 7.44–7.37 (m, 1 H, ArH), 6.96–6.86 (m, 2 H, ArH), 3.88 (s, 3 H, OCH₃), 2.45 (s, 3 H, CH₃).

¹³C NMR (75.4 MHz, CDCl₃): δ = 184.60, 161.31, 134.86, 132.41, 120.45, 110.71, 108.91, 92.27, 87.52, 55.69, 32.62.

MS (EI): *m/z* (%) = 174 (M⁺, 69.00), 159 (100).

Anal. Calcd for C₁₁H₁₀O₂: C, 75.84; H, 5.79. Found: C, 75.82; H, 5.80.

1-(4-Chlorophenyl)undec-2-yn-1-one (2q)

Method A: The reaction of 1-(4-chlorophenyl)undec-2-yn-1-ol (2.7879 g, 10 mmol), Fe(NO₃)₃·9H₂O (201.8 mg, 0.5 mmol), TEMPO (47.1 mg, 0.3 mmol), and NaCl (29.0 mg, 0.5 mmol) in toluene (10 mL) afforded **2q** (2.7184 g, 98%) as an oil.

IR (neat): 2926, 2856, 2237, 2200, 1647, 1587, 1464, 1400, 1261, 1170, 1090, 1013 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 8.06 (dt, *J* = 9.0, 2.1 Hz, 2 H, ArH), 7.44 (dt, *J* = 8.4, 1.8 Hz, 2 H, ArH), 2.49 (t, *J* = 7.2 Hz, 2 H, C≡CCH₂), 1.72–1.60 (m, 2 H), 1.52–1.20 (m, 10 H), 0.88 (t, *J* = 6.6 Hz, 3 H).

¹³C NMR (75.4 MHz, CDCl₃): δ = 176.85, 140.39, 135.30, 130.81, 128.80, 97.48, 79.36, 31.74, 29.07, 28.95, 28.94, 27.72, 22.59, 19.18, 14.04.

MS (EI): *m/z* (%) = 278 [M⁺(³⁷Cl), 0.37], 276 [M⁺(³⁵Cl), 1.27], 139 (100).

HRMS: *m/z* calcd for C₁₇H₂₁³⁵ClO (M⁺): 276.1281; found: 276.1282.

1-(4-(Trifluoromethyl)phenyl)undec-2-yn-1-one (2r)

Method A: The reaction of 1-(trifluoromethyl)undec-2-yn-1-ol (3.1237 g, 10 mmol), Fe(NO₃)₃·9H₂O (202.3 mg, 0.5 mmol), TEMPO (47.2 mg, 0.3 mmol), and NaCl (29.4 mg, 0.5 mmol) in toluene (10 mL) afforded **2r** (3.0551 g, 98%) as an oil.

IR (neat): 2928, 2857, 2237, 2201, 1652, 1583, 1509, 1465, 1411, 1323, 1262, 1172, 1131, 1107, 1065, 1016 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 8.23 (d, *J* = 8.7 Hz, 2 H, ArH), 7.74 (d, *J* = 8.4 Hz, 2 H, ArH), 2.52 (t, *J* = 7.1 Hz, 2 H, C≡CCH₂), 1.75–1.59 (m, 2 H), 1.53–1.22 (m, 10 H), 0.88 (t, *J* = 6.8 Hz, 3 H).

¹³C NMR (75.4 MHz, CDCl₃): δ = 176.91, 139.40 (d, *J* = 1.3 Hz), 134.91 (q, *J* = 32.6 Hz), 129.72, 125.49 (q, *J* = 3.8 Hz), 123.52 (q, *J* = 272.9 Hz), 98.38, 79.38, 31.73, 29.06, 28.94, 27.68, 22.57, 19.17, 13.97.

¹⁹F NMR (282 MHz, CDCl₃): δ = –63.14.

MS (EI): *m/z* (%) = 310 (M⁺, 0.88), 145 (100).

HRMS: *m/z* calcd for C₁₈H₂₁F₃O (M⁺): 310.1545; found: 310.1539.

1-(4-Methoxyphenyl)undec-2-yn-1-one (2s)

Method A: The reaction of 1-(4-methoxyphenyl)undec-2-yn-1-ol (2.7443 g, 10 mmol), Fe(NO₃)₃·9H₂O (202.2 mg, 0.5 mmol), TEMPO (47.0 mg, 0.3 mmol), and NaCl (29.4 mg, 0.5 mmol) in toluene (10 mL) afforded **2s** (2.5087 g, 92%) as an oil.

IR (neat): 2927, 2855, 2237, 2198, 1638, 1596, 1574, 1509, 1463, 1421, 1316, 1253, 1165, 1114, 1029, 911 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 8.10 (d, *J* = 8.7 Hz, 2 H, ArH), 6.94 (d, *J* = 8.7 Hz, 2 H, ArH), 3.88 (s, 3 H, OCH₃), 2.48 (t, *J* = 7.1 Hz, 2 H, C≡CCH₂), 1.73–1.60 (m, 2 H), 1.53–1.20 (m, 10 H), 0.88 (t, *J* = 6.2 Hz, 3 H).

¹³C NMR (75.4 MHz, CDCl₃): δ = 176.84, 164.14, 131.77, 130.22, 113.58, 95.86, 79.52, 55.41, 31.66, 29.00, 28.88, 28.84, 27.73, 22.51, 19.06, 13.96.

MS (EI): *m/z* (%) = 272 (M⁺, 5.74), 135 (100).

HRMS: *m/z* calcd for C₁₈H₂₄O₂ (M⁺): 272.1776; found: 272.1775.

1-(3-Methoxyphenyl)undec-2-yn-1-one (2t)

Method A: The reaction of 1-(3-methoxyphenyl)undec-2-yn-1-ol (2.7443 g, 10 mmol), Fe(NO₃)₃·9H₂O (202.2 mg, 0.5 mmol), TEMPO (46.9 mg, 0.3 mmol), and NaCl (29.1 mg, 0.5 mmol) in toluene (10 mL) afforded **2t** (2.7050 g, 99%) as an oil.

IR (neat): 2925, 2855, 2221, 1645, 1583, 1485, 1463, 1431, 1322, 1271, 1222, 1182, 1104, 1042, 994 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.75 (dt, *J* = 7.5, 1.2 Hz, 1 H, ArH), 7.62 (dd, *J* = 2.7, 1.5 Hz, 1 H, ArH), 7.37 (t, *J* = 7.8 Hz, 1 H, ArH), 7.13 (ddd, *J* = 8.1, 2.7, 0.9 Hz, 1 H, ArH), 3.85 (s, 3 H, OCH₃), 2.48 (t, *J* = 7.2 Hz, 2 H, COCH₂), 1.70–1.60 (m, 2 H), 1.52–1.39 (m, 2 H), 1.39–1.20 (m, 8 H), 0.87 (t, *J* = 6.8 Hz, 3 H, CH₃).

¹³C NMR (75.4 MHz, CDCl₃): δ = 177.95, 159.65, 138.27, 129.44, 122.73, 120.65, 112.81, 96.75, 79.69, 55.38, 31.73, 29.07, 28.96, 28.93, 27.76, 22.58, 19.17, 14.03.

MS (EI): *m/z* (%) = 272 (M⁺, 11.87), 135 (100).

HRMS: *m/z* calcd for C₁₈H₂₄O₂ (M⁺): 272.1776; found: 272.1775.

1-(2-Methoxyphenyl)undec-2-yn-1-one (2u)

Method A: The reaction of 1-(2-methoxyphenyl)undec-2-yn-1-ol (2.7440 g, 10 mmol), Fe(NO₃)₃·9H₂O (202.3 mg, 0.5 mmol), TEMPO (46.9 mg, 0.3 mmol), and NaCl (28.9 mg, 0.5 mmol) in toluene (10 mL) afforded **2u** (2.6250 g, 96%) as an oil.

IR (neat): 2927, 2856, 2212, 1648, 1625, 1596, 1576, 1486, 1464, 1435, 1299, 1237, 1180, 1164, 1132, 1092, 1048, 1023, 908 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 8.01 (dd, *J* = 7.8, 1.8 Hz, 1 H, ArH), 7.53–7.45 (m, 1 H, ArH), 7.04–6.93 (m, 2 H, ArH), 3.90 (s, 3 H, OCH₃), 2.43 (t, *J* = 7.1 Hz, 2 H, COCH₂), 1.67–1.56 (m, 2 H), 1.49–1.19 (m, 10 H), 0.86 (t, *J* = 6.9 Hz, 3 H, CH₃).

¹³C NMR (75.4 MHz, CDCl₃): δ = 177.13, 159.55, 134.60, 132.91, 126.71, 120.03, 112.00, 95.32, 81.66, 55.74, 31.70, 29.04, 28.94, 28.84, 27.75, 22.54, 19.17, 14.00.

MS (EI): *m/z* (%) = 272 (M⁺, 5.59), 135 (100).

HRMS: *m/z* calcd for C₁₈H₂₄O₂ (M⁺): 272.1776; found: 272.1778.

Hexadec-7-yn-6-one (2v)

Method A: The reaction of hexadec-7-yn-6-one (2.3838 g, 10 mmol), Fe(NO₃)₃·9H₂O (202.3 mg, 0.5 mmol), TEMPO (46.8 mg, 0.3 mmol), and NaCl (29.2 mg, 0.5 mmol) in toluene (10 mL) afforded **2v**³¹ (2.3067 g, 98%) as an oil.

IR (neat): 2957, 2932, 2860, 2213, 1675, 1466, 1235 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 2.51 (t, *J* = 7.4 Hz, 2 H, C≡CCH₂), 2.35 (t, *J* = 7.1 Hz, 2 H, COCH₂), 1.71–1.52 (m, 4 H), 1.45–1.20 (m, 14 H), 0.93–0.82 (m, 6 H).

¹³C NMR (75.4 MHz, CDCl₃): δ = 188.62, 94.28, 80.87, 45.48, 31.76, 31.11, 29.07, 28.95, 28.82, 27.68, 23.82, 22.60, 22.35, 18.90, 14.04, 13.84.

MS (EI): *m/z* (%) = 236 (M⁺, 0.55), 221 (M⁺ – CH₃, 0.95), 165 (M⁺ – C₅H₁₁, 42.49), 43 (100).

Anal. Calcd for C₁₆H₂₈O: C, 81.29; H, 11.94. Found: C, 81.61; H, 11.94.

Undec-2-ynal (2w)

Method A: The reaction of undec-2-yn-1-ol (1.6825 g, 10 mmol), Fe(NO₃)₃·9H₂O (202.1 mg, 0.5 mmol), TEMPO (46.9 mg, 0.3 mmol), and NaCl (29.3 mg, 0.5 mmol) in toluene (10 mL) afforded **2w**³² (1.6157 g, 97%) as an oil.

IR (neat): 2927, 2857, 2201, 1670, 1464, 1387, 1137, 1061, 978 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 9.18 (s, 1 H, CHO), 2.41 (t, J = 7.1 Hz, 2 H, $\text{C}\equiv\text{CCH}_2$), 1.65–1.54 (m, 2 H), 1.50–1.20 (m, 10 H), 0.89 (t, J = 6.5 Hz, 3 H).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 177.30, 99.44, 81.66, 31.75, 29.05, 28.94, 28.80, 27.51, 22.59, 19.10, 14.05.

MS (EI): m/z (%) = 165 (M^+ – H, 2.40), 41 (100).

3-Phenylpropynal (2x)

Method A: The reaction of 3-phenylprop-2-yn-1-ol (1.3160 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (203.7 mg, 0.5 mmol), TEMPO (47.3 mg, 0.3 mmol), and NaCl (29.5 mg, 0.5 mmol) in toluene (10 mL) afforded **2x**²² (1.2154 g, 94%) as an oil.

^1H NMR (300 MHz, CDCl_3): δ = 9.43 (s, 1 H), 7.65–7.55 (m, 2 H, ArH), 7.55–7.35 (m, 3 H, ArH).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 176.76, 133.24, 131.26, 128.70, 119.36, 95.08, 88.37.

1-Phenyl-3-phenylprop-2-yn-1-one (2y)

Method A: The reaction of 1-phenyl-3-phenylprop-2-yn-1-ol (2.0825 mg, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.1 mg, 0.5 mmol), TEMPO (47.2 mg, 0.3 mmol), and NaCl (29.4 mg, 0.5 mmol) in toluene (10 mL) afforded **2y**³³ (2.0401 g, 99%) as a solid; mp 49–50 °C (*c*-hexane).

IR (neat): 2198, 1633, 1597, 1579, 1488, 1446, 1315, 1288, 1208, 1170, 1071, 1032, 1012, 995, 934 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 8.26–8.18 (m, 2 H, ArH), 7.71–7.35 (m, 8 H, ArH).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 177.85, 136.70, 134.01, 132.92, 130.69, 129.41, 128.56, 128.50, 119.91, 92.99, 86.75.

MS (EI): m/z (%) = 206 (M^+ , 69.20), 178 (100).

1-Phenylhept-1-yn-3-one (2z)

Method A: The reaction of 1-phenylhept-1-yn-3-ol (1.8789 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (205.9 mg, 0.5 mmol), TEMPO (46.1 mg, 0.3 mmol), and NaCl (28.5 mg, 0.5 mmol) in toluene (10 mL) afforded **2z**²² (1.7268 g, 93%) as an oil.

^1H NMR (300 MHz, CDCl_3): δ = 7.57 (d, J = 7.2 Hz, 2 H, ArH), 7.48–7.34 (m, 3 H, ArH), 2.66 (t, J = 7.5 Hz, 2 H, $\text{C}\equiv\text{CCH}_2$), 1.78–1.64 (m, 2 H), 1.47–1.34 (m, 2 H), 0.94 (t, J = 7.2 Hz, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 188.36, 132.98, 130.59, 128.57, 120.00, 90.48, 87.79, 45.21, 26.19, 22.10, 13.78.

1-Phenylprop-2-yn-1-one (2a)

Method A: The reaction of 1-phenylprop-2-yn-1-ol (1.3214 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.2 mg, 0.5 mmol), TEMPO (47.0 mg, 0.3 mmol), and NaCl (29.5 mg, 0.5 mmol) in toluene (10 mL) afforded **2a**³⁴ (1.2002 g, 92%) as a solid; mp 50–51 °C (*c*-hexane).

IR (neat): 3232, 2093, 1641, 1595, 1579, 1452, 1314, 1247, 1174, 1031, 1005 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 8.17 (d, J = 7.5 Hz, 2 H, ArH), 7.64 (t, J = 7.4 Hz, 1 H, ArH), 7.50 (t, J = 7.7 Hz, 2 H, ArH), 3.45 (s, 1 H, $\text{C}\equiv\text{CH}$).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 177.38, 136.10, 134.52, 129.68, 128.67, 80.76, 80.22.

MS (EI): m/z (%) = 130 (M^+ , 49.22), 102 (100).

Undec-1-yn-3-one (2β)

Method A: The reaction of undec-1-yn-3-ol (1.8227 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (201.8 mg, 0.5 mmol), TEMPO (47.2 mg, 0.3 mmol), and NaCl (29.5 mg, 0.5 mmol) in toluene (10 mL) afforded **2β**²² (1.7275 g, 95%) as an oil.

^1H NMR (300 MHz, CDCl_3): δ = 3.20 (s, 1 H, $\text{C}\equiv\text{CH}$), 2.58 (t, J = 7.4 Hz, 2 H, COCH_2), 1.75–1.60 (m, 2 H), 1.37–1.20 (m, 12 H), 0.88 (t, J = 6.6 Hz, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 187.63, 81.41, 78.21, 45.44, 31.82, 29.33, 29.26, 29.21, 28.86, 23.74, 22.63, 14.08.

1-(4-Trifluoromethylphenyl)hept-2-yn-1-one (2γ)

Method A: The reaction of 1-(4-trifluoromethylphenyl)hept-2-yn-1-ol (2.5566 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (201.3 mg, 0.5 mmol), TEMPO (47.0 mg, 0.3 mmol), and NaCl (29.3 mg, 0.5 mmol) in toluene (10 mL) afforded **2γ**²² (2.2103 g, 87%) as an oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.23 (d, J = 8.1 Hz, 2 H, ArH), 7.73 (d, J = 8.1 Hz, 2 H, ArH), 2.53 (t, J = 6.9 Hz, 2 H, $\text{C}\equiv\text{CCH}_2$), 1.75–1.60 (m, 2 H), 1.58–1.44 (m, 2 H), 0.97 (t, J = 7.2 Hz, 3 H, CH_3).

^{13}C NMR (75.4 MHz, CDCl_3): δ = 176.82, 139.37 (d, J = 1.3 Hz), 134.89 (q, J = 32.6 Hz), 129.70, 125.49 (q, J = 3.8 Hz), 123.51 (q, J = 272.9 Hz), 98.34, 79.38, 29.70, 22.04, 18.90, 13.43.

^{19}F NMR (282 MHz, CDCl_3): δ = –63.23.

1-(Trimethylsilyl)oct-1-yn-3-one (2τ)

Method A: The reaction of 1-(trimethylsilyl)oct-1-yn-3-ol (1.9835 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.3 mg, 0.5 mmol), TEMPO (46.6 mg, 0.3 mmol), and NaCl (29.5 mg, 0.5 mmol) in toluene (10 mL) afforded **2τ**³⁵ (1.8966 g, 97%) as an oil.

IR (neat): 2960, 2151, 1677, 1618, 1583, 1467, 1407, 1252, 1218, 1134, 1083 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 2.55 (t, J = 7.4 Hz, 2 H, COCH_2), 1.72–1.60 (m, 2 H), 1.40–1.20 (m, 4 H), 0.89 (t, J = 6.9 Hz, 3 H), 0.24 [s, 9 H, $\text{Si}(\text{CH}_3)_3$].

^{13}C NMR (75.4 MHz, CDCl_3): δ = 188.12, 102.02, 97.53, 45.23, 31.04, 23.58, 22.33, 13.83, –0.79.

MS (EI): m/z (%) = 197 (M^+ + 1, 0.51), 196 (M^+ , 0.48), 195 (M^+ – 1, 0.55), 181 (M^+ – CH_3 , 14.17), 125 (M^+ – C_5H_{11} , 100).

Anal. Calcd for $\text{C}_{11}\text{H}_{20}\text{OSi}$: C, 67.28; H, 10.27. Found: C, 67.55; H, 10.17.

1-Phenyl-3-(trimethylsilyl)propynone (2a)

Method A: The reaction of 1-phenyl-3-trimethylsilylprop-2-yn-1-ol (2.0434 g, 10 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (202.0 mg, 0.5 mmol), TEMPO (46.8 mg, 0.3 mmol), and NaCl (29.3 mg, 0.5 mmol) in toluene (10 mL) afforded **2a** (1.9876 g, 98%) as an oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.15 (d, J = 7.8 Hz, 2 H, ArH), 7.62 (t, J = 7.4 Hz, 1 H, ArH), 7.49 (t, J = 7.5 Hz, 2 H, ArH), 0.32 [s, 9 H, $\text{Si}(\text{CH}_3)_3$].

^{13}C NMR (75.4 MHz, CDCl_3): δ = 177.78, 136.43, 134.17, 129.64, 128.56, 100.79, 100.57, –0.71.

4-(Trimethylsilyl)but-3-yn-2-one (2ω)

To a 500 mL three-necked flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (8.0907 g, 20 mmol), TEMPO (3.1331 g, 20 mmol), NaCl (1.1752 g, 20 mmol), and toluene (200 mL). The resulting mixture was stirred at r.t. for 10 min, then the atmosphere was replaced by O_2 from an O_2 bag (42 L). After that, 4-(trimethylsilyl)but-3-yn-2-ol (142.3194 g, 1.0 mol) was added to the suspension within 25 min. After the addition, the resulting reaction mixture was stirred at r.t. for an additional 7.4 h with stirring (inside temperature was kept at <55 °C). The reaction was over as monitored by TLC (eluent: PE–EtOAc, 10:1), the resulting mixture was poured into a funnel, the aqueous suspension layer (water with inorganic salts) was separated, and the organic layer was distilled at atmospheric pressure to afford the target compound **2ω**³⁶ [1st fraction: bp 110–120 °C, 144.0454 g, containing 11% of ketone; 2nd fraction: bp 122–148 °C, 76.7192 g, containing 54% of ketone; 3rd fraction: bp 148–158 °C, 73.1452 g, (purity, 96%, which was determined by adding mesitylene as an internal standard)]; combined yield: 90%.

IR (neat): 2965, 2153, 1679, 1421, 1358, 1253, 1198, 979 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 2.33 (s, 3 H, CH₃), 0.23 [s, 9 H, Si(CH₃)₃].

¹³C NMR (75.4 MHz, CDCl₃): δ = 184.44, 102.34, 97.34, 32.47, -0.83.

MS (EI): *m/z* (%) = 140 (M⁺, 2.04), 125 (100).

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Supporting Information for this article is available online at <http://www.thieme-connect.com/ejournals/toc/synthesis>.

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